

1. The Edge Impulse project used raw accelerometer measurements from the three axes: accX, accY, and accZ.
2. The neural network classifier was trained using spectral features extracted from the 3. accelerometer signals. In total, the model used 33 input features, including metrics such as accX RMS, accX skewness, accX kurtosis, accY RMS, and accZ RMS.
3. The classifier produced two possible output classes: nominal and off.
4. The anomaly detection model utilized the same set of spectral features as the neural network classifier. It also operated on 33 features, including accX RMS, accX skewness, accX kurtosis, accY RMS, and accZ RMS.
- 5.

```
off: 0.562500
Anomaly prediction: 16.026260
Starting inferencing in 2 seconds...
Sampling...
Timing: DSP 37 ms, inference 566 us, anomaly 538 us, postprocessing 49 us
#Classification predictions:
  nominal: 0.437500
  off: 0.562500
Anomaly prediction: 16.027981
Starting inferencing in 2 seconds...
Sampling...
Timing: DSP 33 ms, inference 612 us, anomaly 508 us, postprocessing 49 us
#Classification predictions:
  nominal: 0.437500
  off: 0.562500
Anomaly prediction: 16.027740
Starting inferencing in 2 seconds...
Sampling...
Timing: DSP 32 ms, inference 609 us, anomaly 507 us, postprocessing 49 us
#Classification predictions:
  nominal: 0.437500
  off: 0.562500
Anomaly prediction: 16.020761
Starting inferencing in 2 seconds...
Sampling...
```

1. An anomaly threshold of 0.01 was selected because reconstruction errors exceeding this value generally indicated movement patterns that differed from the normal walking behavior used during training.
2. Reconstruction error measures the difference between the original input and the autoencoder's reconstructed output. Higher error values suggest that the observed activity is unfamiliar or anomalous relative to the training data.
3. For successful Arduino deployment, it is necessary to preserve the TensorFlow Lite model, normalization parameters, input dimensions, window size, and anomaly threshold established during model development.

4. Applying the same preprocessing pipeline on the Arduino is essential because the model expects sensor data to be formatted and normalized in the same manner as the data used for training.
5. The anomaly detection system is constrained by the microcontroller's memory and computational resources. Its performance may also be affected by quantization-related accuracy loss, sensor noise, and activities that were not included in the training dataset.

```
Result: Normal activity
Reconstruction error: 0.217215
Result: Normal activity
Reconstruction error: 0.217202
Result: Normal activity
Reconstruction error: 0.217198
Result: Normal activity
Reconstruction error: 0.217203
Result: Normal activity
Reconstruction error: 0.217199
Result: Normal activity
Reconstruction error: 0.217198
```

x